

Simple Alpha Hypotheses for Bybit USDT Perpetuals

Executive summary and hypothesis map

The highest-confidence finding is that crypto does not behave as one homogeneous universe. The evidence is closer to a split market: large, liquid coins show short-horizon continuation more often, while small, illiquid coins show stronger short-term reversal, especially after low-volume or disorderly moves. That distinction appears in cross-sectional crypto research and lines up with the practical microstructure problems of perpetuals: thin books, wick risk, widening spreads, funding distortions, and mark-price liquidations. For a small-account Bybit program, that argues for treating Tier A/B leader continuation and Tier C reversal/fade as different strategy families rather than one breakout template applied everywhere.

1

Continuation evidence is strongest when the asset is already a leader: near prior highs, liquid enough to sustain follow-through, and not yet in a visibly euphoric leverage state. Nearness to the 52-week high predicts future crypto returns in recent academic work, and earlier size/liquidity work finds that “high-momentum” near prior highs is positive for large, liquid coins but negative for small, illiquid ones. That is the cleanest literature-based bridge into a Bybit “risk-on leader breakout long” hypothesis.

2

Short ideas need more confirmation than long ideas. The literature and exchange mechanics both point the same way: liquidation is triggered off mark price, not just last trade, and funding is not a clean directional signal by itself. A spike short should therefore be framed as a post-failure trade, not an anticipatory fade. The strongest candidates are failure after a leverage flush, failed ORB, failed VWAP reclaim, lower-high formation, or price-up/OI-down behavior that suggests short covering rather than fresh trend ignition.

3

Opening-range logic is worth testing in crypto, but the reason is not a literal exchange open. The evidence is that crypto trading activity, volatility, and illiquidity have strong time-of-day seasonality despite 24/7 trading, with major peaks around 16:00–17:00 UTC and broader weekday structure. The correct framing is “synthetic opens where order flow changes,” not “equity-style open transplanted into crypto.” That makes US cash open, funding windows, UTC daily roll, 4-hour candle turns, listing timestamps, and catalyst timestamps the main ORB candidates.

4

The “new perpetuals pump then bleed” idea is plausible, but the direct public evidence for perp launches is still thin relative to spot-listing studies. Spot and cross-listing studies show positive abnormal returns around listings and meaningful reversal or underperformance afterward, while exchange product design has increasingly added pre-market and innovation-zone style leverage around new instruments. That combination is enough to justify a formal event study, but not enough to treat launch-window shorting as already proven. It should start as a research priority with strict survivorship controls and exchange-specific stratification.

5

For regime definition, internal market variables rank above external sentiment. BTC/ETH trend state, cross-sectional breadth, alt-risk appetite, and a derivatives composite built from price, OI, funding, and liquidation context are the best first-layer regime features because they are available in near real time and directly tied to perp execution. External series such as Fear & Greed, Google Trends, or social sentiment are lower priority because they are either BTC-heavy, publication-lagged, or noisy. ⁶

The first tests should not be the most exotic ones. The order should be: leader breakout continuation, weak-asset spike fade, risk-off exhaustion spike short, ORB failure around funding or US cash-open windows, and only then newly listed perp launch shorts. That ordering best balances plausibility, simplicity, small-account suitability, and the ability to reject ideas quickly without exotic data engineering. ⁷

Mapping the five core hypotheses

Hypothesis	Evidence for	Evidence against	Best regime	Worst regime	Required data	Proposed test design
Risk-on leader breakout long	Large and liquid coins show short-term momentum; nearness to prior highs predicts future crypto returns; intraday momentum exists in crypto and changes with liquidity and jump conditions. ⁸	Crypto momentum is crash-prone; results weaken outside leaders and large caps; euphoric leverage can turn continuation into snapback. ⁹	Risk-on trend expansion, majors-led or sector-led breadth, positive but not extreme funding, rising OI with intact depth. ¹⁰	Late euphoric melt-up, distressed illiquidity, listing mania, negative breadth divergence. ¹¹	Last/trade, mark, index, funding, OI, fees, depth, spread, breadth, prior-high distance. ¹²	Cross-section rank Bybit perps daily: top decile relative strength, within 0%–8% of 20d/60d highs, broad market breadth filter, then compare daily-close breakout vs 5m/15m/30m ORB continuation entries.

Hypothesis	Evidence for	Evidence against	Best regime	Worst regime	Required data	Proposed test design
Risk-off exhaustion spike short	Leverage flushes often reset funding and can create better later entries; low-quality assets reverse more; pump-and-dump type spikes are prone to quick reversals. ¹³	Genuine squeeze ignition, legal/access catalyst, or sector rotation can keep running; shorting before backside confirmation is structurally dangerous in perps because liquidation uses mark price. ¹⁴	Risk-off trend down, negative breadth, weak spot participation, crowded short-cover pop, poor liquidity. ¹⁵	Fresh narrative ignition, exchange listing, token-specific access news, broad market upside expansion. ¹⁶	Mark/last/index, OI, funding, liquidation prints, depth/spread, VWAP, listing/catalyst metadata. ¹⁷	Event bars where price spikes x ATR in risk-off regime; enter only after failed ORB, VWAP loss, lower high, or price-up/OI-down state; compare time-stop and origin-fill exits.
Weak-asset spike fade in risk-on	Small/illiquid names and names far from prior highs are the part of crypto where reversal dominates; one-week "high-momentum" is negative for small/illiquid coins. ¹⁸	In strong alt-rotation, laggards can catch up instead of fading; some spikes are genuine discovery after prolonged neglect. ¹⁹	Risk-on but majors-led, weak alt breadth, local pump in structurally weak coin, thin-book Tier C. ²⁰	Broad altseason, sector ignition, exchange-backed listing news, improving liquidity. ²¹	Relative performance vs BTC/ETH/sector, prior-high distance, spread/depth, OI/funding, VWAP, wick metrics. ²²	Define weakness by underperformance, poor prior-high proximity, low trend quality, and weak liquidity; wait for failed continuation trigger; test tight stop at high vs VWAP reclaim with wider targets to VWAP/20 EMA/origin.

Hypothesis	Evidence for	Evidence against	Best regime	Worst regime	Required data	Proposed test design
Newly listed perps that pump then bleed	Listing studies show positive abnormal returns around exchange listings with later reversal; recent exchange design adds pre-market and innovation-zone leverage, which can amplify launch dislocation. ²³	Direct public event-study evidence for Bybit/Binance/OKX/Hyperliquid perp launches remains limited; outcomes likely heterogeneous by spot availability, narrative, token age, and exchange design. ²⁴	Neutral or weak market, pre-listing run-up, thin initial book, no strong spot discovery anchor, post-airdrop/unlock overhang. ²⁵	Strong bull regime, major CEX spot support, broad narrative leadership, low float with sustained rotation. ²⁶	Exchange announcement timestamps, onboard/list/launchTime, first-trade data, mark/index, OI/funding, order book, delists. ²⁷	Event study from 2020 onward by venue: returns at 5m/15m/30m/1h/4h/24h/3d/7d/14d/30d; stratify by token age, pre-spot availability, pre-run-up, market regime, and pre-market vs direct launch. Test launch ORB, VWAP loss, day-1 close failure, first lower high.
ORB and opening-range failure in 24/7 crypto	Crypto shows strong time-of-day seasonality in volume, volatility, and liquidity; weekday commonality strengthened around ETF era; session boundaries plausibly matter through order-flow clustering. ²⁸	There is little direct crypto-specific ORB literature proving profitability; the effect may be just seasonality rather than a breakout edge. ²⁹	US cash open, UTC day open, 4h turns, funding windows, listing timestamps, catalyst timestamps. ³⁰	Flat overnight liquidity, weekends without catalysts, structurally wide spreads, random chop. ³¹	Intraday trade/mark/index, VWAP, depth/spread, funding windows, exchange clock, DST-safe calendars, OI/funding. ³²	Test continuation and failure variants across 1m/3m/5m/15m/30m/60m ranges, by open definition; require VWAP and regime filters; include mark-price stop simulation and spread/slippage by tier.

Regime measurement and activation map

The most useful definition of “market mood” for a Bybit perp program is not a headline sentiment series. It is a layered state model built from internal market structure first, then external proxies only as secondary context. This is partly because perps now dominate crypto derivatives activity and often dominate price discovery, and partly because funding alone is a trailing or mixed signal rather than a pure directional read. Coinbase’s institutional primer explicitly argues that funding is often trailing price and more useful as a volatility or positioning indicator than as a standalone directional one. ³³

A ranked proxy stack for testing is straightforward. First comes BTC/ETH trend plus cross-sectional breadth: BTC/ETH above or below intermediate moving averages, slope of those averages, realized volatility, percent of Bybit linear perps above 20d and 60d averages, percent making 20d or 60d highs, and advance/decline or cross-sectional momentum breadth. These are fully measurable from exchange data, have no publication lag beyond exchange timestamps, and map directly to the “risk-on leader breakout” versus “fade weak spike” split. The strength of prior-high proximity in crypto returns makes breadth-plus-nearness especially attractive as a first-pass regime layer. ³⁴

Second comes a derivatives composite, but only as an interaction term with price. The useful states are not “funding high” or “OI high.” They are combinations such as price up/OI up with moderate funding, price up/OI flat with extreme funding, price up/OI down after a squeeze, price down/OI down in a flush, and basis/funding dislocations when arbitrage capacity is weakening. Recent academic and practitioner evidence supports this caution: funding skews positive by construction on many venues, liquidations can mechanically reset funding, and in stress episodes the pass-through from basis to funding can weaken when OI collapses. ³⁵

Third comes alt-risk appetite. ETHBTC, SOLBTC, BTC dominance, alt breadth, and sector breadth matter because the same local price pattern means different things in different cross-sections. A clean breakout in a leader during majors-led strength is not the same as a low-quality laggard pump during a narrow BTC-led bounce. Coinbase’s June 2026 positioning note is useful here because it shows a majors-led recovery can coexist with positive BTC/ETH funding and rising OI while altcoin OI dominance remains historically depressed. That is exactly the kind of state where breakout longs should concentrate in leaders and weak-asset fades should remain live. ³⁶

Fourth comes liquidity state: spread, displayed depth, turnover, and volume shock. This matters more on Bybit than many spot-only studies imply because fills, wick exposure, and stop execution quality are first-order determinants of realized edge for small accounts. Microstructure research and exchange-level liquidity work both show that liquidity and adverse selection conditions are time-varying and predictive, while Kaiko’s recent work on Bybit after the 2025 hack is a reminder that venue liquidity itself can change materially around exchange-specific events. ³⁷

Fifth comes session and order-flow state. The strongest empirical support is that crypto has persistent intraday seasonality in activity, volatility, and illiquidity, so a session label is not cosmetic. Use this as an activation variable for ORB/failure ideas, not as a general directional bias. ²⁸

External proxies rank below these. Alternative.me’s Fear & Greed is transparent about being primarily Bitcoin-focused, which makes it acceptable as a coarse overlay but not as a direct alt-perp regime switch. Google Trends and social sentiment have some documented association with crypto returns and volatility,

but the evidence is mixed and practical implementation suffers from revision, timing, and taxonomy problems. ETF flow proxies are useful for BTC/ETH regime annotation, but only after careful timestamp alignment. ³⁸

Regime classification and strategy activation

Regime	How to operationalize	Active	Reduced	Disabled
Risk-on trend expansion	BTC and ETH above rising 20d/60d averages; breadth improving; leaders near highs; price up/OI up; funding positive but not extreme. ³⁹	Leader breakout longs; post-catalyst continuation; ORB continuation in leaders.	Weak-asset fades only when clear structural laggard.	Anticipatory parabolic shorts.
Risk-on late euphoric	Breadth high but narrowing; funding elevated or capped; large gap from VWAP/EMA; liquidation clusters; repeated extension bars. ⁴⁰	Failed breakout shorts, backside-confirmed blowoff shorts.	Leader breakout longs.	Fresh breakout chasing in weak liquidity.
Risk-off trend down	BTC/ETH below falling MAs; negative breadth; price down/OI up or flat; rallies fail under VWAP. ⁴¹	Risk-off exhaustion spike shorts; failed ORB shorts; crowded-long unwind shorts.	Liquidation-flush reversal longs in majors only.	Breakout longs outside event-driven leaders.
Risk-off squeeze-prone	Negative broad trend but extreme negative funding or recent leverage flush; price down/OI down; liquidations just cleared. ⁴²	Wait for squeeze then failure; short only after backside confirmation.	Reversal longs with very short holding windows.	Blind spike shorts into active squeeze.
Choppy/range	Flat BTC/ETH slopes, mixed breadth, muted OI trend, low dispersion.	Mean-reversion and ORB failure strategies.	Continuation breakouts.	Listing-window aggression.
Distressed illiquidity	Median spreads widening, depth deteriorating, exchange stress or venue event. ⁴³	Only smallest-risk reversal/failure tests; maybe majors only.	Most alt strategies.	Small-cap breakout longs and large size execution.

Regime	How to operationalize	Active	Reduced	Disabled
Sector-specific ignition	Neutral broad market but sector breadth and one or two leaders break out with OI support.	Sector leader breakouts; catalyst continuation bases.	Weak-asset fades in the ignited sector.	Broad-market risk-off shorts in that sector.
High-volume synthetic open	UTC day open, US cash open, London overlap, 4h turn, or catalyst window with above-baseline volume/volatility. ⁴⁴	ORB continuation or failure depending on regime.	Slower swing entries during the same window.	Random discretionary entries without open definition.
Funding-window rebalance	Around contract funding settlement or immediate post-settlement interval; volume/vol shift and order rebalancing. ⁴⁵	Funding-window ORB/failure; crowded-long unwind.	Standard breakout signals without timing edge.	Strategies that assume a single universal 8h schedule.
New-listing launch	First trade, first auction release, or conversion from pre-market to live perp. ⁴⁶	Listing ORB, listing VWAP-loss short, first-lower-high short.	Generic trend-following templates.	Any contract requiring long history.

Opening ranges in twenty-four seven crypto

The case for ORB-style testing in crypto is not the existence of a formal exchange open. It is the existence of repeated order-flow transitions. The strongest academic evidence is that returns, trading activity, volatility, and illiquidity show pronounced and surprisingly synchronized intraday patterns across crypto exchanges and pairs. Trading activity and illiquidity rise into a clear peak around 16:00–17:00 UTC, and returns are weakest in early UTC morning and stronger later in the day. A separate 2024/2025 study finds BTC and ETH intraday seasonality and commonality that is strongest on weekdays and altered by the US spot-Bitcoin ETF era. ⁴

That implies five practical “opens” are worth first-pass testing on Bybit. The first is the UTC daily candle open, because it is universal, stable, and used by many crypto participants. The second is the 4-hour candle boundary, because many crypto traders anchor risk and narrative structure there; this is not proven in the literature, but it is sufficiently widespread and simple to test. The third is the funding-session boundary, which is exchange-specific and must be read from instrument metadata rather than assumed fixed, because Bybit and Binance both allow varying funding intervals and can modify them for individual contracts. The fourth is the US equity cash open, because that is where offshore perps increasingly intersect with traditional macro and ETF-related flow. The fifth is the event-specific open: listing time, auction end, or known catalyst timestamp. ⁴⁷

CME matters less as a classical “reopen” than it once did because CME’s crypto futures are now effectively 24/7 except for short maintenance windows, so the better modern test is “CME maintenance-adjacent flow” rather than “CME open.” For US cash open, use actual market calendars and convert 9:30 a.m. ET to UTC

historically, rather than hard-coding a constant offset. The same applies to London or Asia session definitions if used. ⁴⁸

The highest-priority ORB variants are not symmetric. Continuation should be tested mainly in leaders and during risk-on trend expansion; failure/fade should be tested in weak assets, risk-off regimes, and listing windows. A practical starting grid is opening-range lengths of 1m, 3m, 5m, 15m, 30m, and 60m for event opens, and 5m, 15m, 30m, 60m for session opens. Range length should interact with asset tier: shorter windows for BTC/ETH and high-liquidity names, longer windows for Tier C where noise and spread risk are larger. ⁴⁹

Stops should be validated on mark-aware logic, not just traded-price bars. Bybit, Binance, OKX, and Hyperliquid all use mark price in liquidation or margin control, and Bybit and Binance explicitly distinguish last price from mark price for liquidation and unrealized PnL. That means a backtest using only trade-bar highs and lows can overstate survivability for both longs and shorts. The first stop family to test is opposite side of opening range plus a small ATR cap. The second is VWAP reclaim/loss. The third is a mark-price liquidation buffer for leverage-aware sizing. ⁵⁰

Targets should remain simple: 1R/2R/3R, session high/low extension, VWAP reversion for fade trades, and time stops. Given the cost profile of small-account perp trading, an ORB edge that only survives with cherry-picked trailing exits is more likely to be overfit than real. The minimal bar for acceptance is that the edge survives maker/taker assumptions, widening spreads around the chosen open, funding during holds that cross settlement windows, and outlier wicks. ⁵¹

Research on the directional hypothesis families

Risk-on leader breakouts

The cleanest literature-supported version of this idea is not “buy any breakout in a good tape.” It is “buy strong, liquid leaders near prior highs when the market is risk-on but not yet euphoric.” Two strands of evidence support this. First, recent cross-sectional work finds that nearness to the 52-week high is positively associated with subsequent crypto returns. Second, size-and-liquidity work finds positive short-horizon momentum and positive prior-high-style “high-momentum” for large, liquid coins, while the same signal flips negative in small, illiquid names. ²

That matters for Bybit because official exchange mechanics show that perps track spot through funding and mark/index design while providing leveraged, shortable access that dominates trading volume in practice. Inference is unavoidable here: the strongest published factor evidence is mostly coin-level rather than Bybit-perp-specific, but Coinbase and theory papers both support the idea that perp prices are close enough to spot for cross-sectional return structure to matter, while execution and liquidation mechanics remain distinctly perp-native. ⁵²

The best regime filters are narrow and mechanical. Require BTC and ETH above rising intermediate averages, positive cross-sectional breadth, positive but not extreme funding, and OI rising at a moderated pace. Exclude names with vertical last-24h distance from VWAP or 20 EMA beyond a chosen percentile, because these are closer to euphoric chase conditions than clean breakouts. Coinbase's June 2026 note is helpful here: it explicitly describes a majors-led recovery driven more by leverage than by broad spot

conviction, with depressed altcoin OI dominance. In such states, breakout continuation should be concentrated in leaders and not generalized to the full alt universe. ³⁶

The timing question is whether ORB helps. My reading of the evidence is that ORB is worth testing, but not assumed superior a priori. Intraday crypto does have momentum and fluid regime dependence, yet the literature does not directly prove that ORB outperforms close-based breakout entry in perps. So the correct contract is comparative: test daily close breakout, 4h close breakout, and ORB continuation with 5m/15m/30m ranges on the same leader subset. Accept ORB only if it reduces false starts or improves fill-adjusted expectancy. ⁵³

A falsifiable Bybit contract is therefore simple: long only top-quartile relative-strength perps within 5%–8% of 20d or 60d highs; activate only when breadth and BTC/ETH trend are positive; enter on either daily close breakout or session/listing-neutral ORB break that also holds above session VWAP; invalidate on OR low or VWAP loss; exit on 2R, 3R, or close below 10 EMA/1-day low. Reject if returns are carried by a handful of trend months or if performance collapses once names with wide spread or thin depth are excluded. ⁵⁴

Risk-off exhaustion spike shorts

The core mechanism here is not “mean reversion because market down.” It is that in a negative or fragile tape, upside spikes are often driven by short covering, thin-book vacuum, or reflexive leverage clearing rather than durable demand. Glassnode’s leverage-flush work is useful because it shows that large deleveraging events can reset funding sharply and change the structure of positioning. The academic liquidity-provision literature complements that by showing stronger reversal in lower-quality or low-activity names. ⁵⁵

The way this fails is equally important. When a spike coincides with true information—new exchange access, legal clarification, major listing, sector reclassification, or meaningful spot-led rotation—fading it can be fatal. That is why the trigger should be backside confirmation, not raw extension. This aligns with the user’s own constraint against anticipatory parabolic shorts and with exchange liquidation design: mark-price liquidations create asymmetric danger when one tries to pick an exact top. ¹⁴

Among confirmation triggers, the highest-confidence set is the simplest. First, loss of session or event VWAP after the spike. Second, failed ORB: price breaks the opening range high, returns inside the range, then breaks the range low. Third, lower high after the spike. Fourth, price-up/OI-down or flat OI, which is more consistent with shorts being forced out than with durable fresh long demand. Fifth, spread and depth deterioration during the push, which hints at vacuum rather than sponsorship. I treat these as strong research inferences from the combined literature and exchange mechanics, not as already-proven standalone academic rules. ⁵⁶

A clean test design is event-based. In a risk-off regime, flag all bars where a perp rises more than 2–3 intraday ATR from prior close or event anchor. Split them into price-up/OI-up, price-up/OI-flat, and price-up/OI-down cases. Enter short only on one of three failure signatures: VWAP loss, failed ORB, or lower-high break. Compare exits at VWAP touch, pre-spike origin, and 2R/3R. Reject if the edge vanishes once tokens with contemporaneous exchange announcements or known catalysts are removed. ⁵⁷

Weak-asset spike fades in risk-on markets

This is the strongest short-side hypothesis in the whole set. The reason is that it is directly supported by the best size/liquidity evidence in crypto. Small and illiquid coins show one-week reversal, and one-week prior-high or “high-momentum” signals are negative for those names while positive for large and liquid coins. A weak asset spiking in a risk-on tape is therefore not automatically “late catch-up”; often it is exactly the part of the universe where reversals dominate. ¹⁸

The difficult part is separating genuine laggard catch-up from terminal pump/fade. The best definitions of weakness are relative, not absolute: underperformance versus BTC, ETH, and sector leaders; large distance from 20d/60d highs; poor trend quality in the prior weeks; weak liquidity persistence; and a tokenomic overhang such as imminent unlock or recent airdrop distribution if those timestamps are available. This is one area where the idea likely belongs more in Tier C than Tier A/B, because the continuation literature is strongest in large/liquid names and the reversal literature is strongest in the smaller or lower-quality tail. ⁵⁸

The payoff design should be asymmetric by construction. If this hypothesis is real, it will often produce many small losses and fewer large winners. That argues for tight invalidation—spike high, OR high, or VWAP reclaim—and wider targets: VWAP, 20 EMA, base midpoint, or pre-spike origin. It should be tested with hard time stops because weak names can simply go dead after the pump and stop offering follow-through. ⁵⁹

Synthetic-session ORB failure is particularly attractive here because it creates a tight stop and a simple rejection rule. Use only on names already classified as weak by relative strength and prior-high distance. If the broad market is risk-on but alt breadth is mediocre and the local spike occurs in a laggard, then failed ORB plus VWAP loss is a much cleaner design than a raw fade above resistance. ⁶⁰

Newly listed perpetuals

The broad claim that “new perps pump then bleed” is plausible, but today it is still more hypothesis than established fact. The strongest direct listing literature is about spot and exchange cross-listings, where studies find positive abnormal returns around listings and later reversal or underperformance effects. RockawayX’s multi-CEX work also shows a common pattern of pre-listing run-up and softer median path afterward for venues including Bybit. That is enough to motivate a perp launch event study, but not enough to skip the study and greenlight the strategy. ⁶¹

The reason to take the hypothesis seriously anyway is exchange design. Bybit now has pre-market perpetuals and innovation-zone products with distinct fees and launch mechanics. OKX runs both standard perpetual listings and pre-market futures that can convert into normal contracts. Binance publishes precise launch timestamps and onboard dates for USD[Ⓢ]-M futures. Hyperliquid’s listing regime materially changes with HIP-3 builder-deployed perpetuals in 2026, including independent deployer settings and open-interest caps. Those design features can create leverage shocks, low-float discovery bursts, and later liquidity decay. ⁶²

A proper event study should therefore be exchange-specific and era-specific. For Bybit, use announcement pages plus instrument `launchTime`; for Binance, announcement launch timestamps plus `onboardDate`; for OKX, listing notices and `listTime` /instrument updates; for Hyperliquid, use historical market

metadata and, from the HIP-3 era onward, classify launches by deployer structure. Historical market-data vendors such as Tardis are useful because they provide instrument-level histories, order books, liquidations, mark-price channels, and exchange/state updates for Bybit, Binance, OKX, and Hyperliquid across materially long samples. ⁶³

The event windows should be fixed before seeing results: first 5m, 15m, 30m, 1h, 4h, 24h, 3d, 7d, 14d, and 30d. Stratify by market regime, token age, pre-listing spot availability, pre-listing run-up, narrative sector, market-cap/liquidity tier, presence of airdrop or unlock overhang, and whether the launch began as pre-market trading. For trade design, compare four simple shorts: immediate launch-window failure, first loss of launch VWAP, day-one close failure below launch range midpoint, and first lower high after day one. Also test the opposite—launch ORB continuation—because some launches do trend cleanly in strong regimes. ⁶⁴

The biggest implementation risks are survivorship and lifecycle handling. Instruments that delist, rebase, or trade briefly are part of the phenomenon and cannot be excluded. Bybit, Binance, OKX, and Hyperliquid all publish delisting behaviors that can forcibly settle or cancel positions, and OKX explicitly notes that parameters such as funding, leverage, and limit price may be adjusted before delisting. Any new-listing strategy that ignores this will overstate returns. ⁶⁵

Additional simple alpha ideas and concrete strategy contracts

Additional simple hypotheses worth testing

Idea	Economic mechanism	Required universe	Entry and confirmation	Stop and exit	Required data	Main failure mode
Funding-window ORB failure	Position rebalancing around funding can create transient directional pushes and quick failures. ⁴⁵	BTC/ETH plus liquid alts	Define 5m/15m range around funding settlement; take opposite-side break only after first break fails back into range.	Stop outside first break extreme; exit VWAP or 2R.	Intraday bar, session VWAP, funding timestamp, spread.	No true flow shift when funding interval changed or too small to matter.

Idea	Economic mechanism	Required universe	Entry and confirmation	Stop and exit	Required data	Main failure mode
Liquidation-flush rebound long	Forced selling can overshoot fair value, especially after leverage resets and funding normalizes. ⁴²	BTC/ETH and liquid alts only	Large downside move with OI down and long-liquidation burst, then reclaim of flush VWAP or OR high.	Stop at flush low or mark-buffer under it; exit VWAP extension / 2R / time stop.	Liquidations, OI, mark/last, VWAP.	Falling knife in true regime break.
Crowded-long unwind short	Positive funding and rising OI into stalled price can leave longs vulnerable to air pockets. ⁶⁶	Leaders and high-beta majors	Price stalls near highs while funding high and OI elevated; enter on failed breakout or VWAP loss.	Stop above failed high; exit 20 EMA / range low / 2R.	Funding, OI, price, breadth.	Strong trend resumes and carries high funding for longer than expected.
Failed sector rotation short	Sector-wide chase into second- and third-line names often fades when leadership narrows. ⁶⁷	Sector baskets	Weak sector member spikes while sector leader stalls or reverses; enter on failed ORB or relative-strength rollover.	Stop above local high; exit sector midpoint or pre-spike base.	Sector classification, relative strength, intraday bars.	Genuine broad sector ignition.
Post-catalyst continuation base long	Information underreaction can cause pause-then-go behavior in leaders after real catalysts. ⁶⁸	Major listed names with identifiable catalysts	After catalyst impulse, enter on tight base breakout above VWAP with stable OI and non-extreme funding.	Stop below base low; exit 2R/3R or close under 10 EMA.	Catalyst timestamps, OI, funding, VWAP, depth.	Event already fully priced or liquidity too thin.

Idea	Economic mechanism	Required universe	Entry and confirmation	Stop and exit	Required data	Main failure mode
Daily open reversal	Crypto's UTC day change may concentrate positioning resets and chart anchoring. ⁶⁹	Liquid alts and majors	Gap/impulse away from UTC open then reclaim through open and VWAP.	Stop beyond extension extreme; exit to opposite session extreme.	UTC timestamps, VWAP, bar data.	Trend day overwhelms reversion.
Cross-coin lead/lag within sectors	Delayed response by smaller or less liquid names to leader moves can create follow or fade setups. ⁷⁰	Sector pairs or leader/laggard sets	Leader breaks or fails first; laggard enters only once relative spread crosses threshold.	Stop on spread reversion through entry; exit after fixed lag window.	Pair alignment, intraday returns, sector map.	Relationship unstable across regimes, fees dominate.

Concrete strategy contracts

The point of these contracts is not to recommend live deployment. It is to reduce each idea to something a backtesting team can falsify fast.

Leader breakout long contract. Universe: top 30–50 Bybit linear USDT perps by rolling 30-day median dollar volume, plus a secondary Tier C sleeve for separate analysis. Filter: BTC and ETH above rising 20d averages; at least 55%–65% of chosen universe above 20d average; asset in top 20% relative strength over 20d; asset within 0%–8% of 60d high. Entry: either daily close above prior 20d high, or 15m ORB break above range high while above session VWAP. Confirmation: OI change positive but below 90th percentile of recent breakouts; funding positive but below the asset's 80th or 90th percentile. Stop: OR low or 1.0–1.5 ATR, whichever is tighter. Target: 2R, 3R, or exit on close below 10 EMA. Required data: trade and mark bars, funding, OI, breadth, fees, spread estimate. Reject if edge only appears without cost modeling or only in 2020–2021 type mania. ⁷¹

Weak-asset spike fade contract. Universe: Tier B/C Bybit perps with tradable minimum depth thresholds. Filter: broad market risk-on by BTC/ETH and breadth, but asset underperforms BTC and sector leader over 20d, sits more than 15% below 60d high, and ranks in worse half of trend-quality metrics. Event: intraday spike greater than 2 ATR or 8%–15%, depending on tier. Entry: first failed continuation trigger, defined as break above local range high that closes back below it, plus session VWAP loss. Confirmation: OI flat or down over entry window, funding rising but not yet extreme, spread widening relative to trailing median. Stop: spike high or VWAP reclaim. Target: session VWAP, 20 EMA, pre-spike origin, or 3R. Reject if slippage and wick-adjusted losses dominate any large-winner profile. ⁷²

Risk-off exhaustion spike short contract. Universe: all liquid and semi-liquid Bybit perps, excluding active fresh listings and major catalyst names. Filter: BTC below falling 20d/60d averages or breadth below 40%; asset not a sector leader. Event: price spikes >2 ATR in 4h or >1.5 ATR intraday against regime. Entry: after lower high, failed 5m/15m ORB, or decisive VWAP loss. Confirmation: price-up/OI-down preferred; if OI up, require faster failure. Stop: above spike extreme plus small mark buffer. Target: pre-spike midpoint then origin; optional scale at VWAP. Reject if best results come from entering before failure confirmation. 73

Funding-window ORB failure contract. Universe: top-liquidity Bybit USDT perps whose instrument metadata confirms funding interval. Open definition: the 5m, 15m, and 30m windows beginning at funding settlement and the immediately following bar. Entry: take opposite-side range break only after the first break fails and price crosses VWAP. Filters: only in either trendless/choppy regimes or crowded-long/crowded-short states. Stop: failed-break extreme. Target: opposite side of range, then 2R. Reject if returns vanish after including spread expansion around settlement windows. 74

Listing VWAP-loss short contract. Universe: all Bybit linear launches from 2020 onward, with Binance/OKX/Hyperliquid as research comparators. Event anchor: official launch time or auction end time. Entry: no trade in first X minutes; go short only after price loses launch VWAP and fails one reclaim attempt, or after day-one close below launch-range midpoint. Stops: launch high for day-one; failed-reclaim high for intraday. Targets: 24h close, 3d hold, and origin fill. Filters: stronger if pre-listing run-up large, spot already available elsewhere, market regime neutral-to-weak, and early depth thin. Reject if effect disappears once delisted and failed instruments are restored to sample. 75

US cash-open ORB continuation/failure contract. Universe: BTC, ETH, SOL, and top liquid beta alts. Open definition: first 5m/15m/30m after 9:30 a.m. ET converted to historical UTC via calendar, not static offset. Continuation version: long/short with range break only when aligned with broad regime and above/below VWAP. Failure version: take the opposite side only after the first break re-enters the range. Stop: opposite side of range with ATR cap. Target: 1R/2R/3R or close by fixed intraday cutoff. Reject if performance is no better than a simple same-time momentum baseline. 76

Priorities for the backtesting team

The first research block should be the leader breakout long. It has the cleanest literature support, the simplest regime filters, and the least dependence on rare event metadata. It also diversifies a Qullamaggie-style continuation program without forcing low-liquidity names into a template where academic evidence is weakest. The first rejection test should be brutal: does the edge survive using mark-aware stops, taker fees at realistic rates, and a slippage model that worsens around breakouts? 77

The second block should be weak-asset spike fades. This idea fits small accounts better than many momentum ideas because capacity is not the bottleneck, and the literature specifically says the tail of the universe is where reversal lives. The first rejection test here is execution realism: require minimum displayed depth, add bad-wick scenarios, and compare last-price versus mark-based stop hits. If the idea needs fantasy fills at the exact spike high, discard it. 78

The third block should be risk-off exhaustion spike shorts. It overlaps with the weak-asset theme but adds an explicit broad-market regime dimension. Its main challenge is false positives from real catalysts. So the first version should exclude listing days, pre-market conversion days, and major venue announcement windows, then add them back later as separate strata. 79

The fourth block should be ORB failure around funding and US cash-open windows. This should be treated as a timing enhancer, not a standalone belief system. The primary question is whether synthetic opens add information after regime and asset-quality filters are already in place. If not, ORB should be dropped rather than protected with ever more conditions. ⁸⁰

The fifth block should be new-perp listing studies. This is strategically important, but it is fourth or fifth in sequence because the direct public evidence remains thinner and the data engineering burden is higher. The project should still begin early enough that delisted instruments and announcement metadata are archived before gaps appear. ⁸¹

Research limits and source notes

The biggest limitation is direct empirical evidence on Bybit perp launches and on crypto-specific ORB profitability. There is good evidence for crypto intraday seasonality, cross-sectional momentum versus reversal splits, listing effects in crypto more broadly, and perp microstructure mechanics. There is much less public, high-quality work that directly says “Bybit linear USDT perpetuals launched at time t systematically bleed from day 1 to day 7” or “funding-window ORB failure is profitable after costs.” Those are valid research targets precisely because the literature is incomplete. ⁸²

A second limitation is that many academic studies use coin-level or spot-heavy datasets, not executable perp data. That means OHLCV-only results should remain screening evidence. The live research stack must eventually distinguish last price, mark price, index price, funding, OI, fees, spread, depth, liquidations, contract status, and delisting behavior. Exchange documentation strongly supports this requirement. ⁸³

A third limitation is instrument lifecycle bias. Bybit, Binance, OKX, and Hyperliquid all have explicit delisting or settlement processes, and recent Hyperliquid HIP-3 changes mean listing mechanics there are no longer stationary through time. Any historical event study that silently drops dead markets or ignores regime changes in listing rules will be mis-specified. ⁸⁴

Source list and notes on quality

Bybit official documentation. Best source for launch timestamps, funding interval history, open interest endpoints, mark-price design, liquidation mechanics, fee schedule, innovation-zone and pre-market mechanics, and delisting rules. High quality for execution and lifecycle details. ⁸⁵

Binance official documentation. Strong comparative reference for launch timestamps, mark price, funding intervals, open interest endpoints, and delisting conventions. High quality for cross-venue robustness checks. ⁸⁶

OKX official documentation. Useful for comparative work on funding, mark price, listing notices, pre-market futures, and delisting/process adjustments. High quality for lifecycle handling. ⁸⁷

Hyperliquid official documentation. Necessary for comparative DEX references, especially because listing mechanics changed materially under HIP-3 and because open-interest caps and deployer settings can affect launch behavior. High quality for mechanism details, but historically shorter data sample. ⁸⁸

Tardis historical data documentation. High-value infrastructure source for verifying multi-venue historical availability, captured channels, and coverage dates for trades, books, liquidations, mark price, OI, and funding across Bybit, Binance, OKX, and Hyperliquid. Very useful for event-study feasibility. ⁸⁹

Coinbase Institutional research. Good practitioner source on perpetual futures mechanics and the nontrivial interpretation of funding rates, including the warning that funding is often trailing price. Strong quality, but still practitioner rather than peer-reviewed. ⁹⁰

Glassnode research. Good practitioner source on leverage flushes, funding resets, and liquidation context. Strong for mechanism intuition and regime annotation, less suitable than exchange data for precise backtests. ⁹¹

Brauneis et al. on intraday seasonality. Best academic source here for the claim that crypto has strong, synchronized time-of-day patterns in trading activity, volatility, and illiquidity despite 24/7 trading. High quality and directly relevant to ORB research framing. ⁹²

Wen et al. on intraday predictability. Useful academic support that crypto has both intraday momentum and reversal, with behavior contingent on liquidity, jumps, and macro announcement conditions. Relevant mainly for timing research, not direct strategy proof. ⁹³

Bianchi and Dickerson on liquidity provision. High-quality academic support for reversal concentrating in low-volume, low-quality, smaller, less liquid, and more volatile crypto pairs. Central source for weak-asset reversal and fade hypotheses. ⁹⁴

Fičura on size, volume, momentum, and reversal. Very important bridge source for separating large/liquid continuation from small/illiquid reversal, including prior-high-style signal asymmetry. High practical relevance to tiered-perp research. ⁹⁵

Jia et al. on nearness to the 52-week high. Strong recent evidence that prior-high proximity matters in crypto cross-sections. Useful for leader-breakout screening. ⁹⁶

Grobys et al. on momentum crashes. Useful caution against over-reading momentum evidence without crash control and volatility management. Important for rejection rules. ⁹⁷

Listing-effect research. The academic exchange-listing paper and the comparative CEX research note are the best available public references for listing event direction, but both are broad crypto listing studies rather than perp-launch studies. Useful, but indirect. ⁶¹

Recent research on limits to arbitrage in perps. Useful for understanding why OI collapses and funding become less informative under stress, and for avoiding naive use of funding as a signal. Quality is promising but newer and less battle-tested than the older cross-sectional literature. ⁹⁸

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